



ATS Short Course 2025

Reactive Transport



Knoxville, TN and virtual
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ATS Short Course 2025

Reactive Transport



Sergi Molins

BERKELEY LAB



IDEAS
Watersheds

This work is supported by the U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research under Award no. DE-AC02-05CH11231.





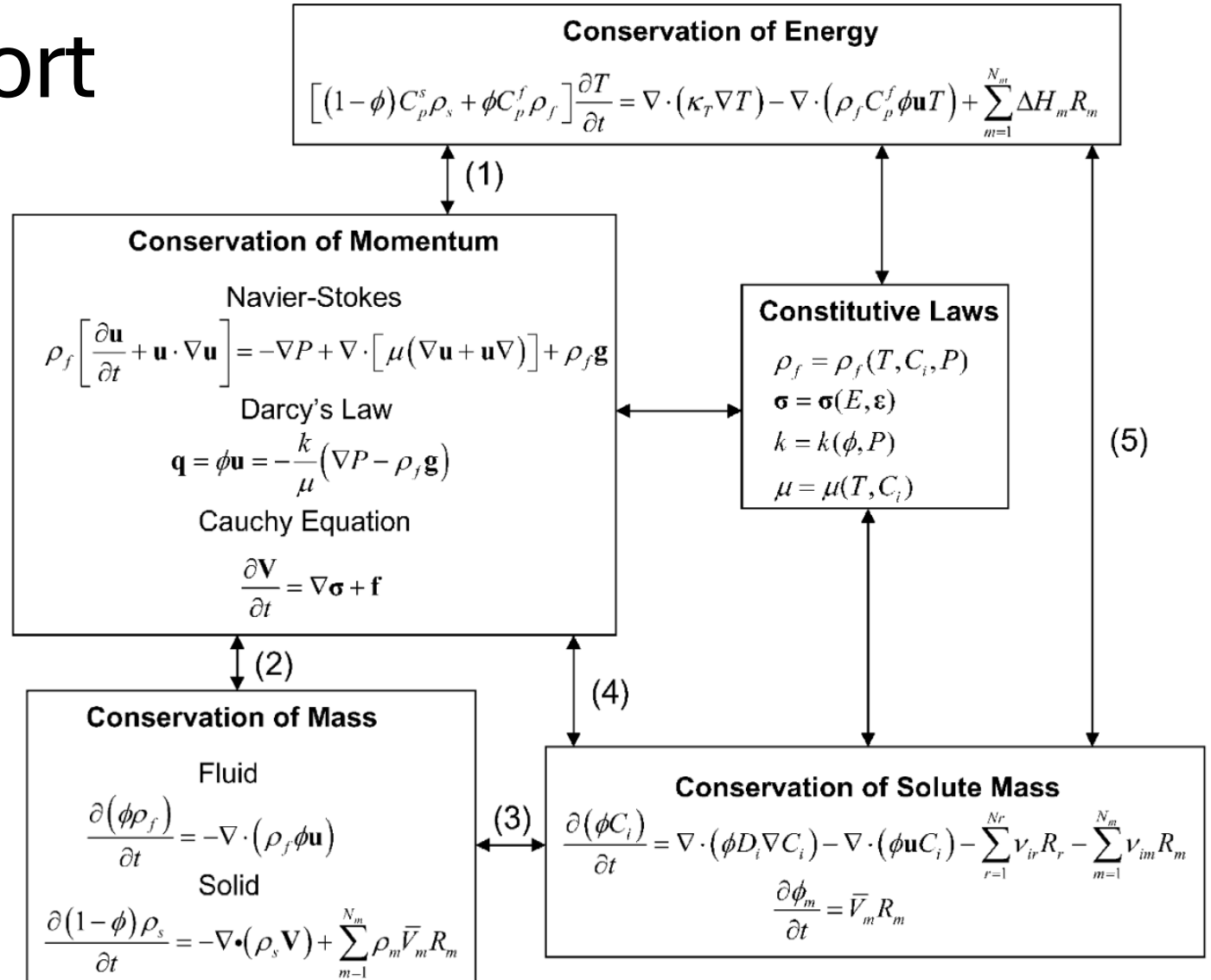
Session Outline

- Introduction
 - Reactive transport
 - Reactive transport with ATS
- Hands-on exercises
 - Exercise 1: Simple single-component rate
 - Exercise 2: Multicomponent geochemistry



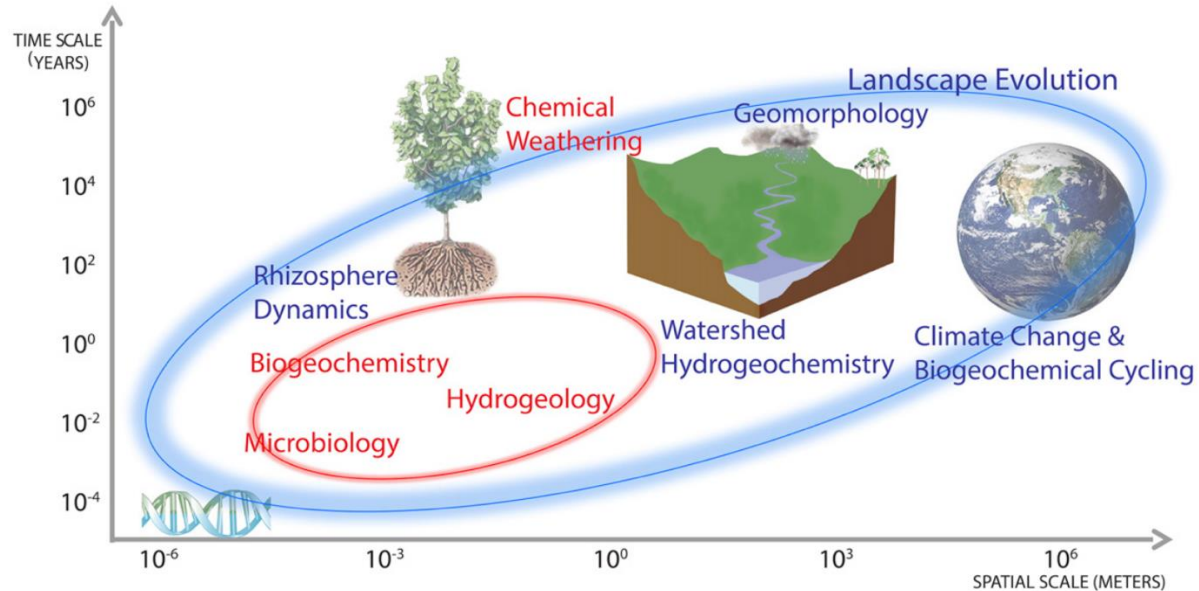
Reactive Transport

A tool for the analysis of coupled physical, chemical, and biological processes





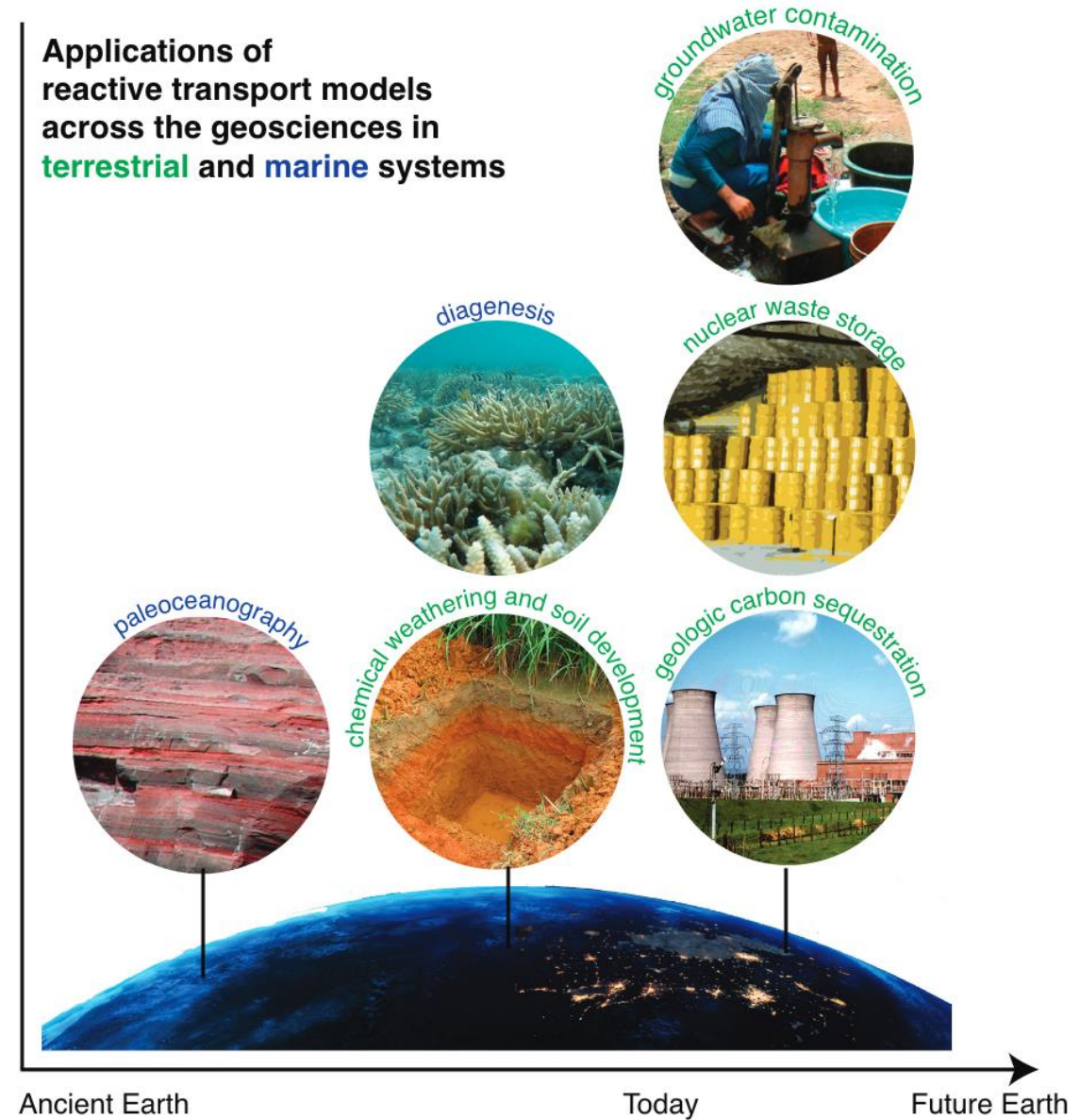
Reactive Transport



L. Li et al. / Earth-Science Reviews 165 (2017) 280–301

“There is a pressing need to couple RTMs with other community models to explore non-linear interactions among the atmosphere, hydrosphere, biosphere, and geosphere.”

Applications of reactive transport models across the geosciences in **terrestrial** and **marine** systems

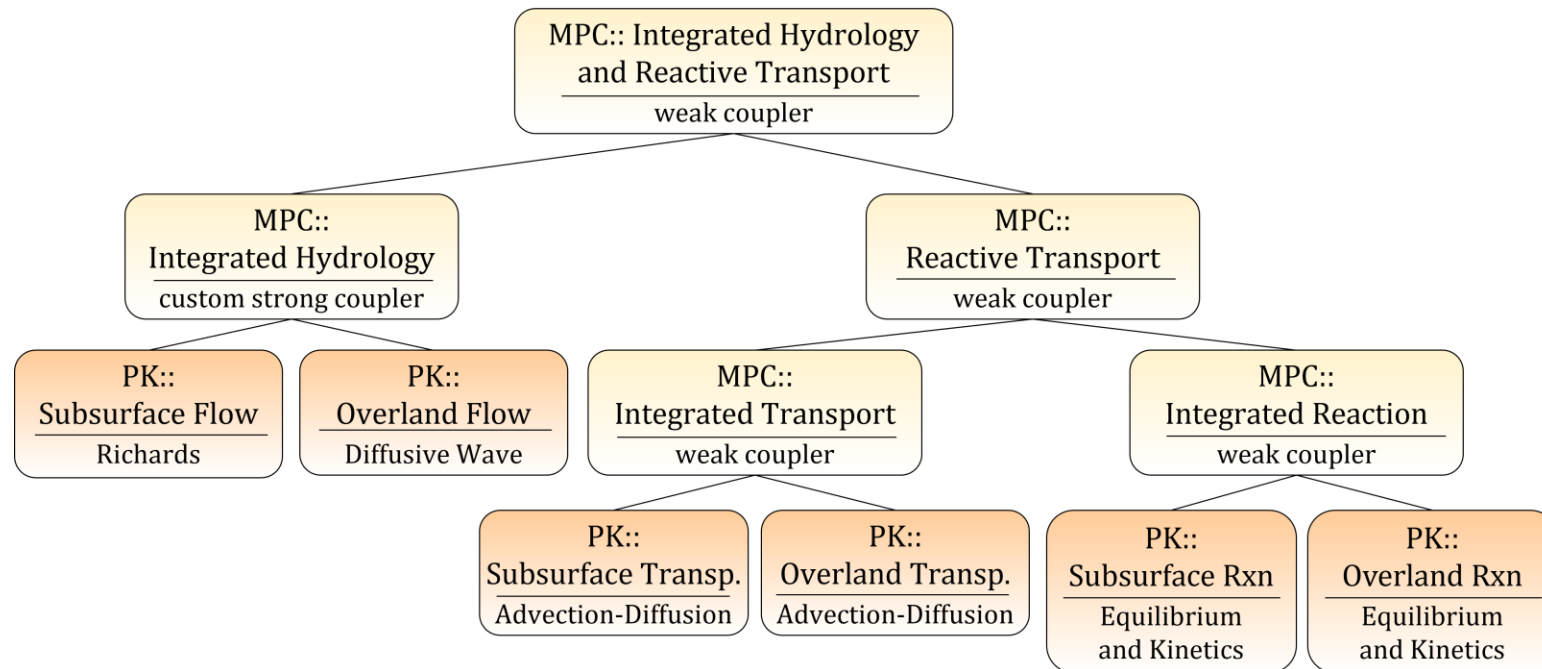


Maher and Mayer / Elements 15 (2019) 81–86



Reactive Transport with ATS

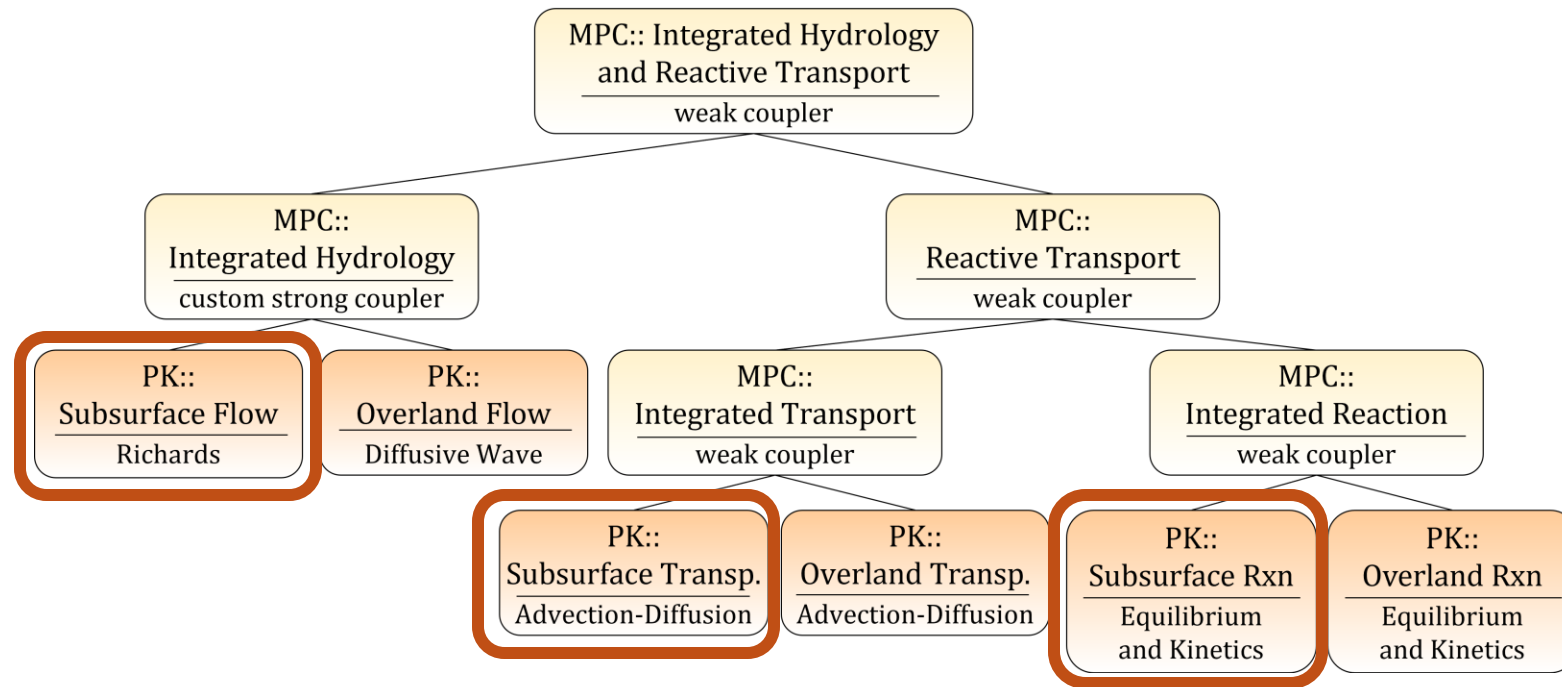
- Unique distributed process-based modeling capability for integrated hydrology problems & watershed hydrogeochemistry





Reactive Transport with ATS

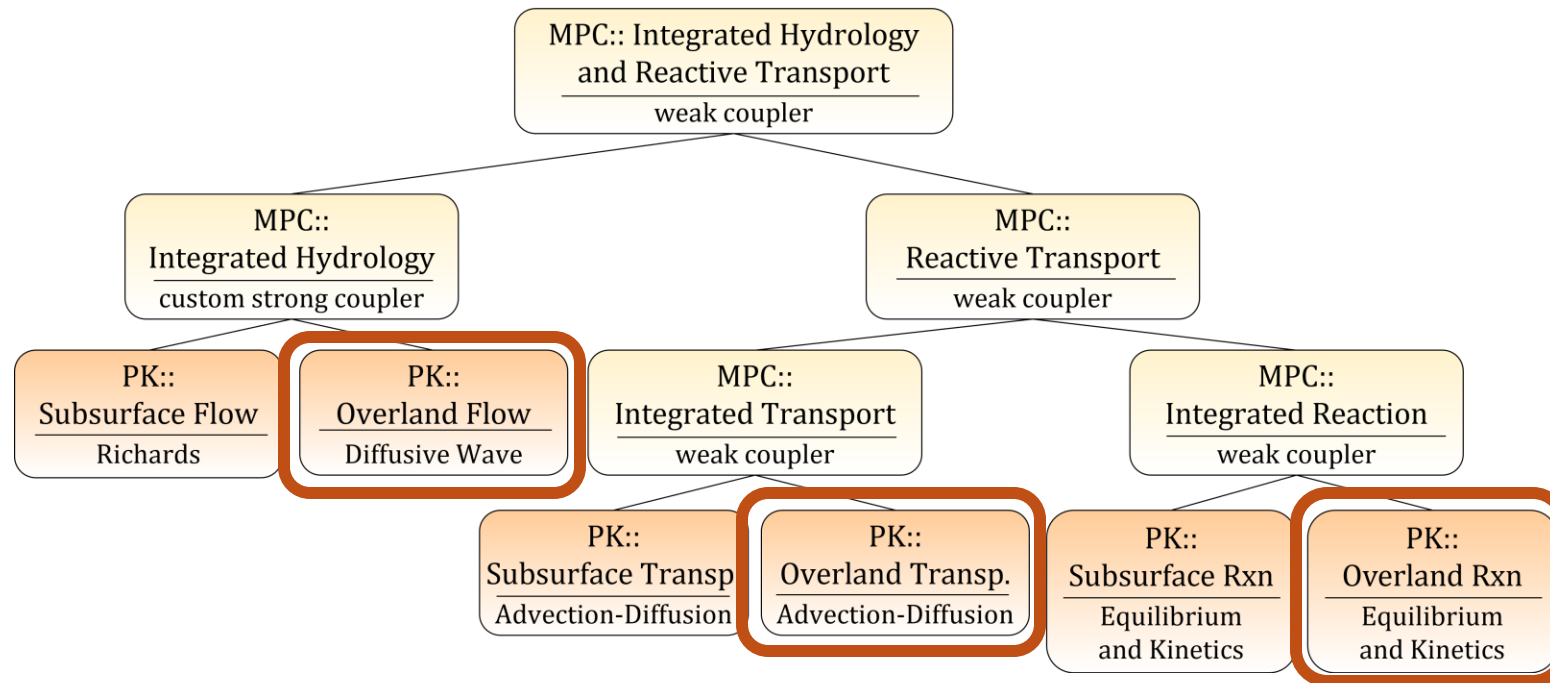
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Reactive Transport with ATS

- Unique distributed process-based modeling capability for integrated hydrology problems & watershed hydrogeochemistry



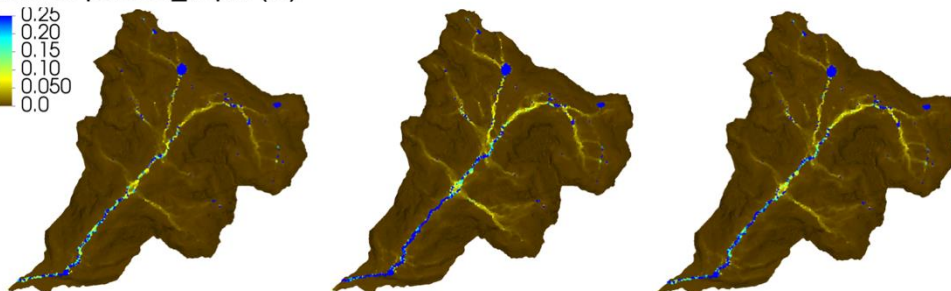
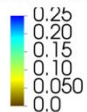


Application: Copper Creek

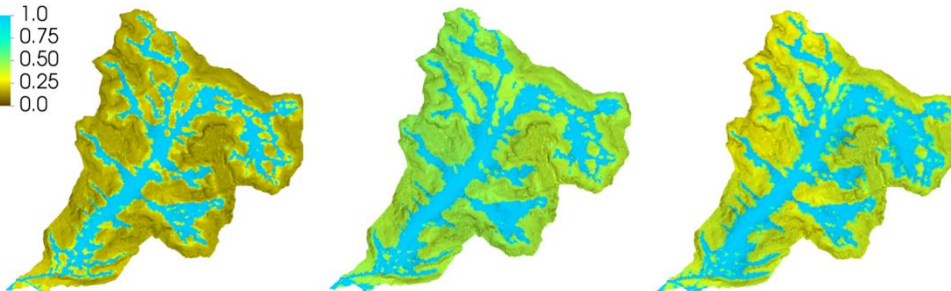
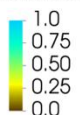
Snow water equivalent (m)



Surface ponded_depth (m)



Subsurface saturation



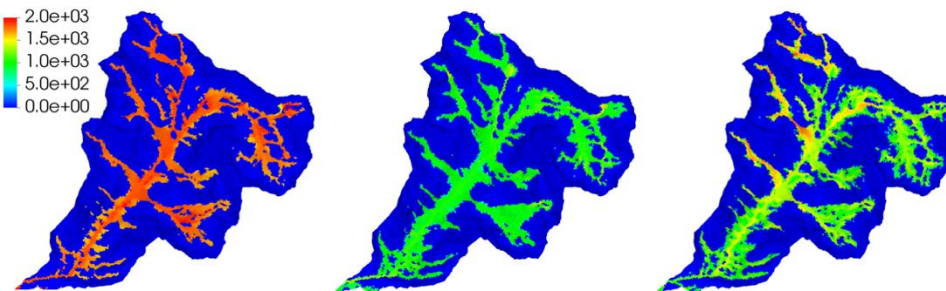
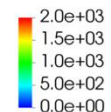
Jan 2017

May 2017

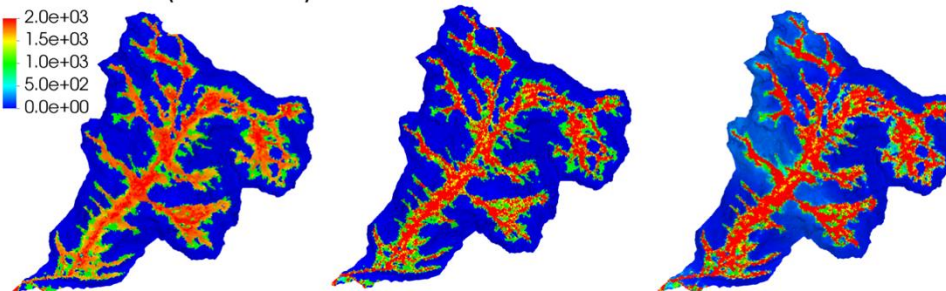
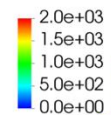
July 2017

Xu et al. (2022) Water Resour. Res.

Surface Ca Concentration



Ca Concentration (Groundwater)



Jan 2017

May 2017

July 2017

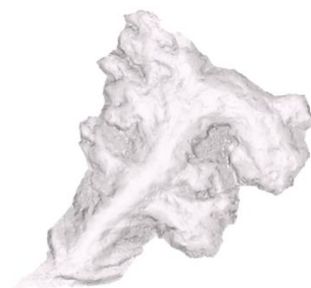
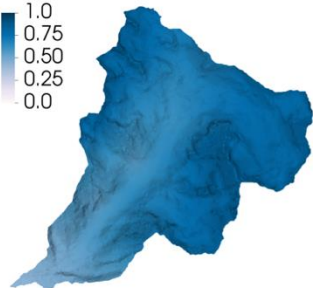


Application: Copper Creek

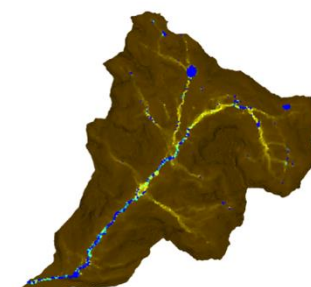
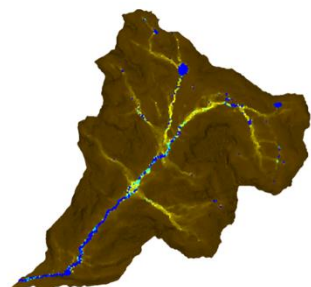
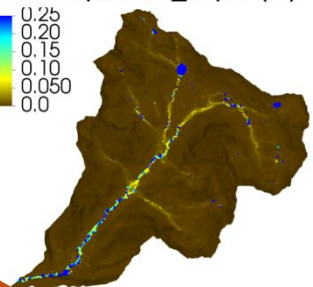
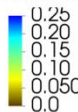
surface

Xu et al. (2022) Water Resour. Res.

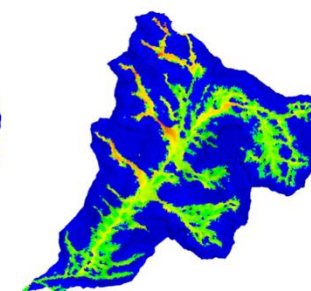
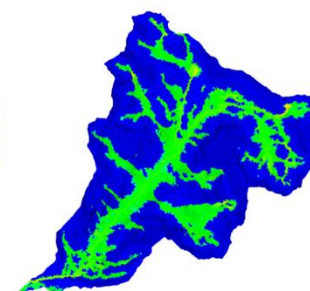
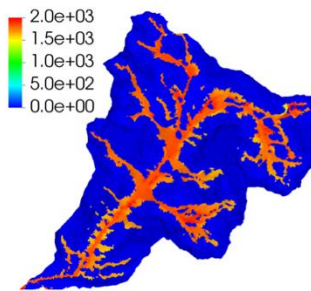
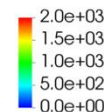
Snow water equivalent (m)



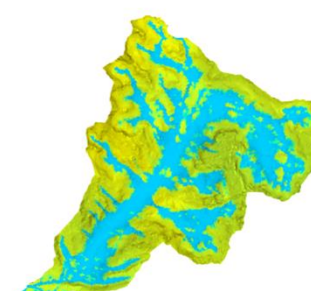
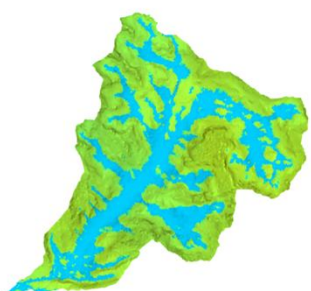
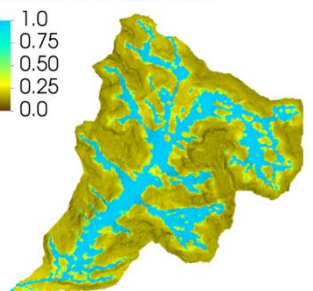
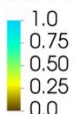
Surface ponded_depth (m)



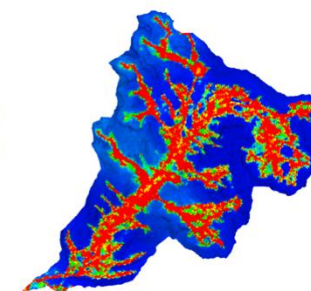
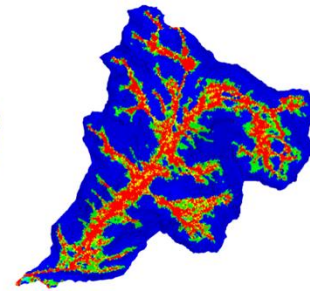
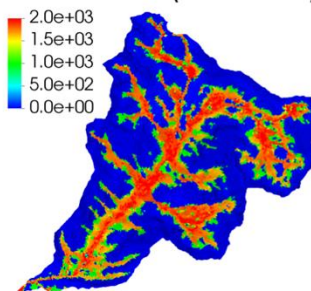
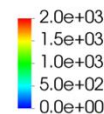
Surface Ca Concentration



Subsurface saturation



Ca Concentration (Groundwater)



Jan 2017

May 2017

July 2017

Jan 2017

May 2017

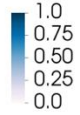
July 2017



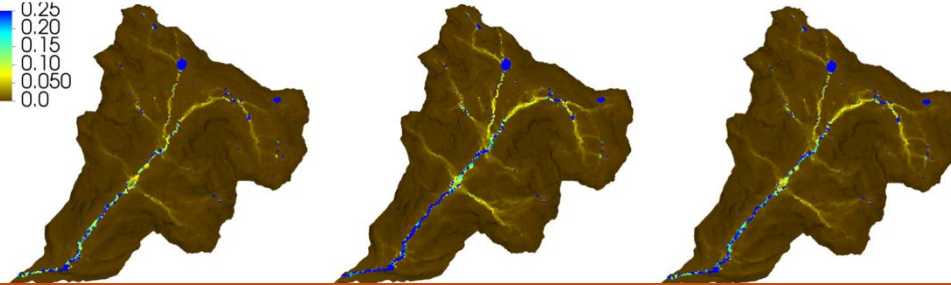
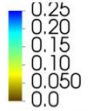
Application: Copper Creek

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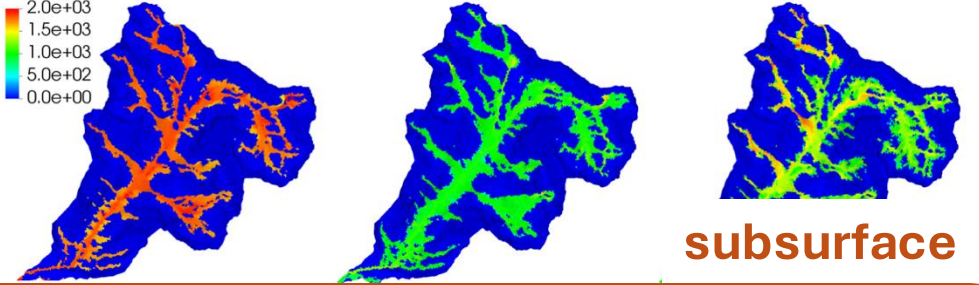
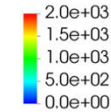
Snow water equivalent (m)



Surface ponded_depth (m)

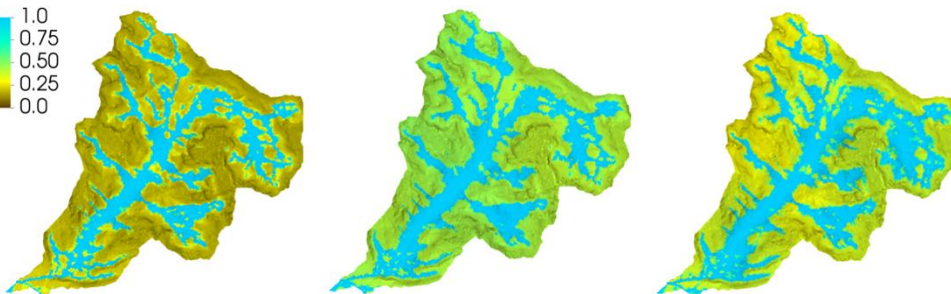
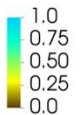


Surface Ca Concentration



subsurface

Subsurface saturation

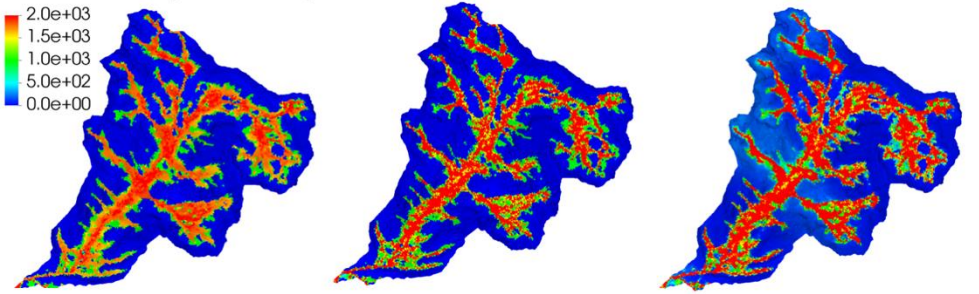
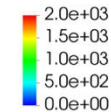


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Ca Concentration (Groundwater)



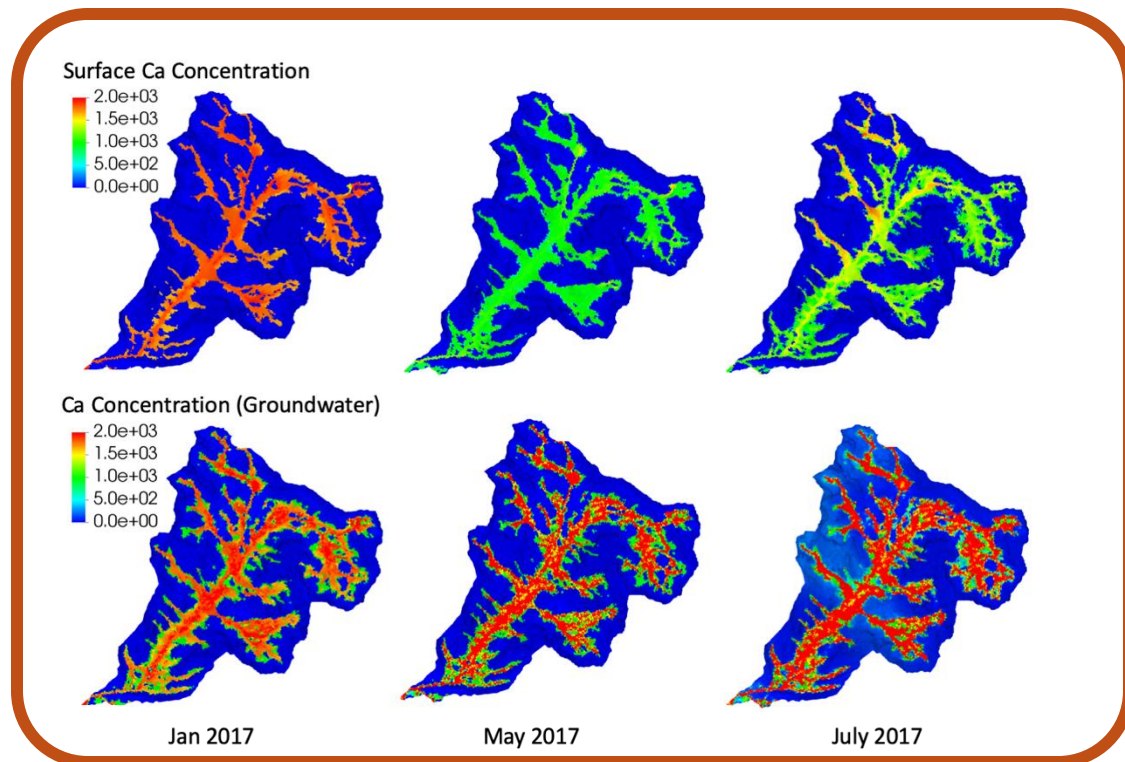
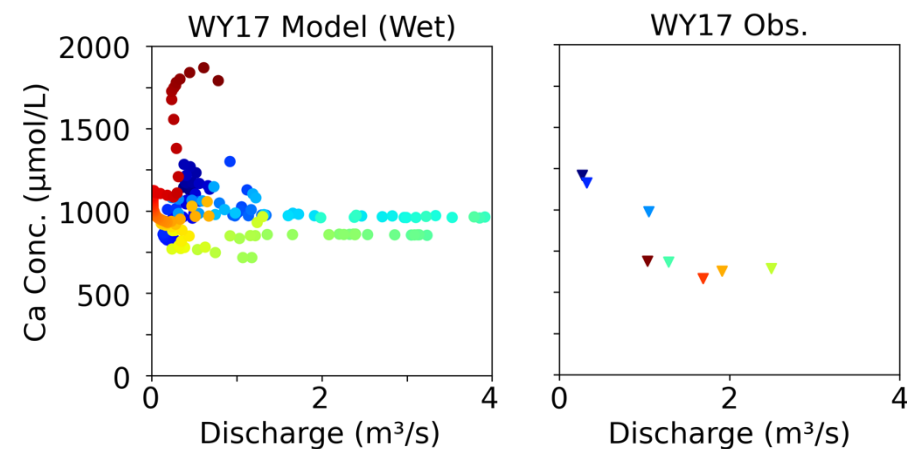
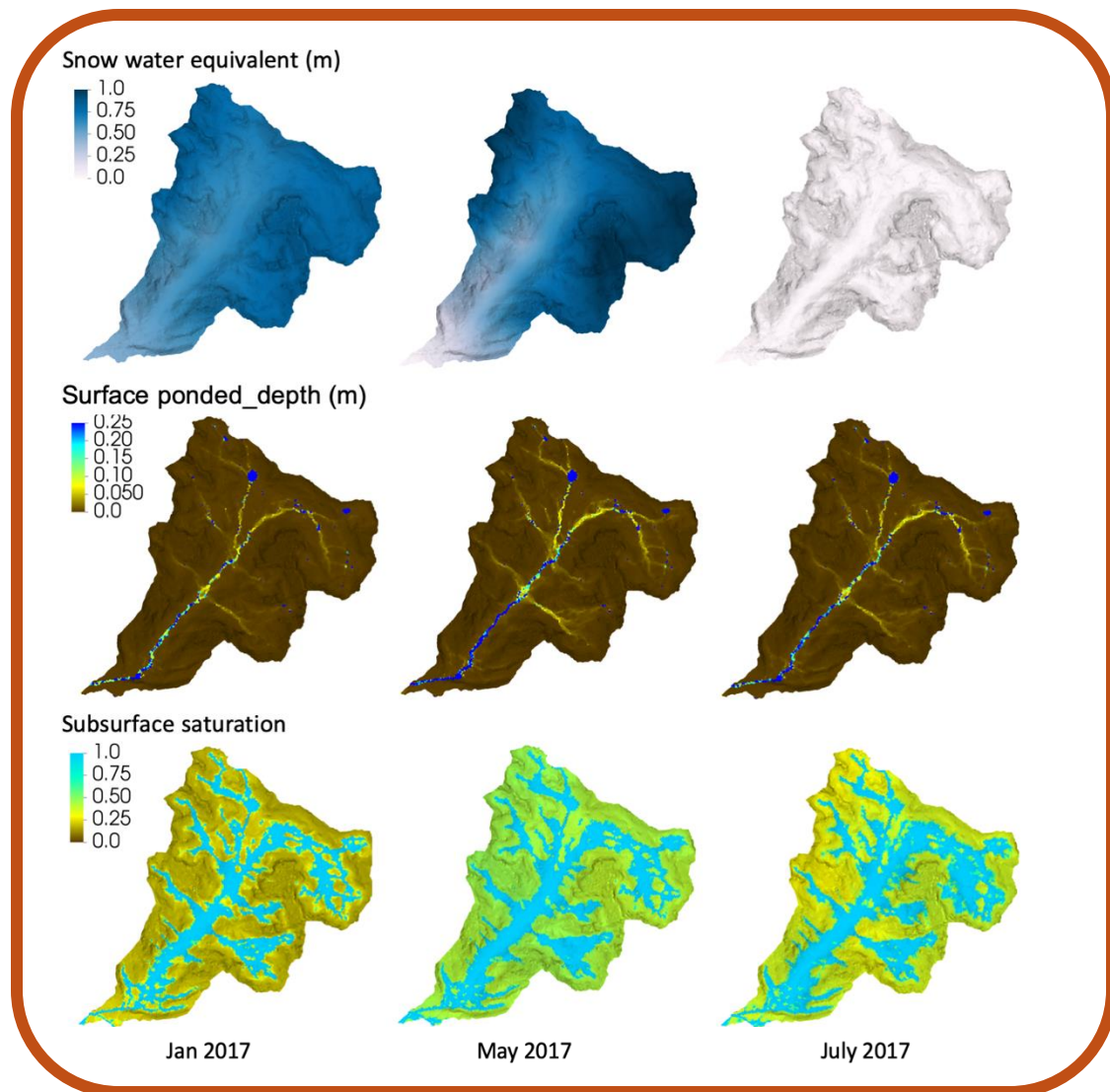
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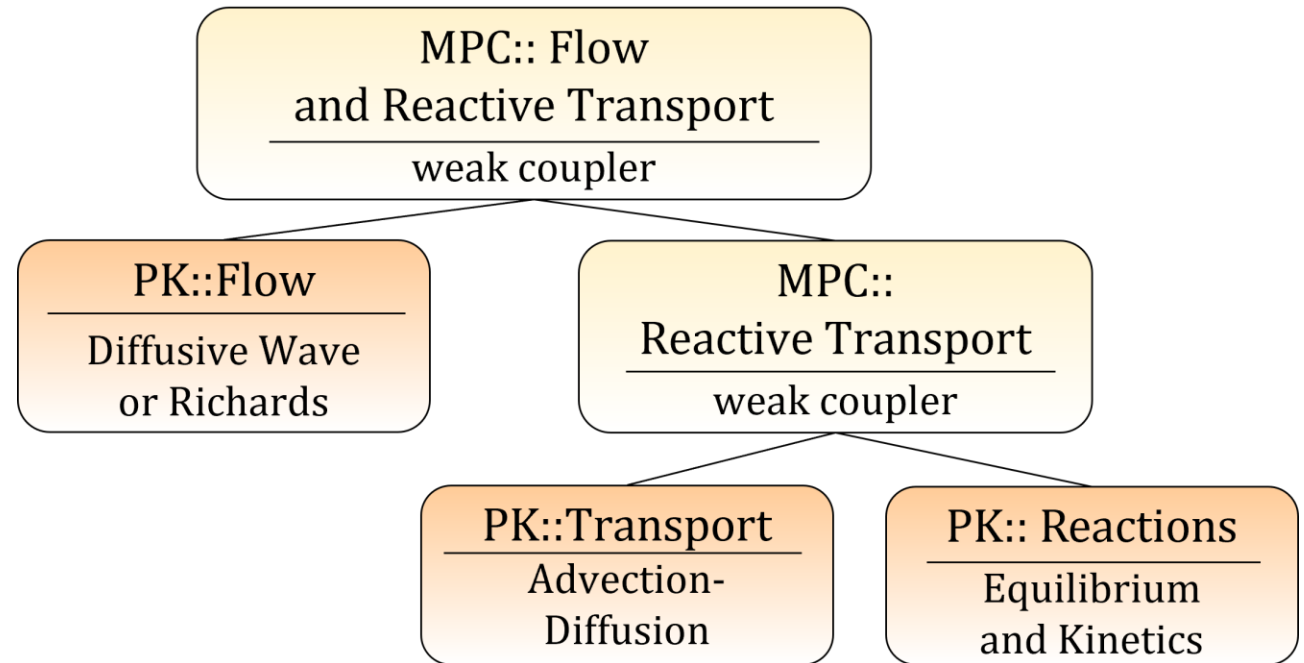


Application: Copper Creek



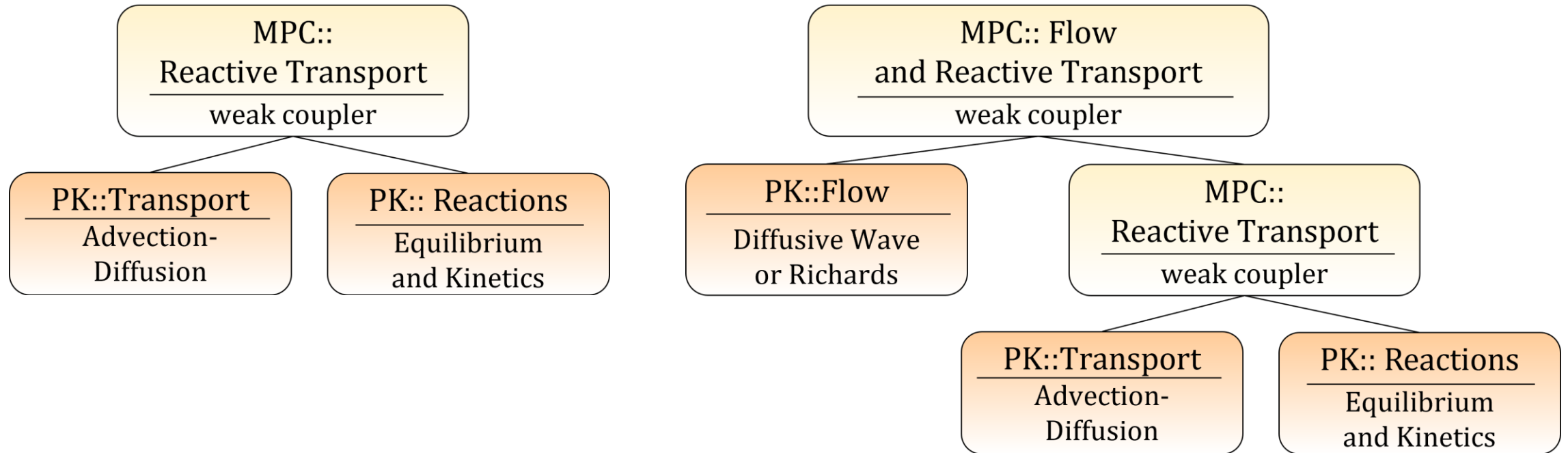


Reactive Transport with ATS



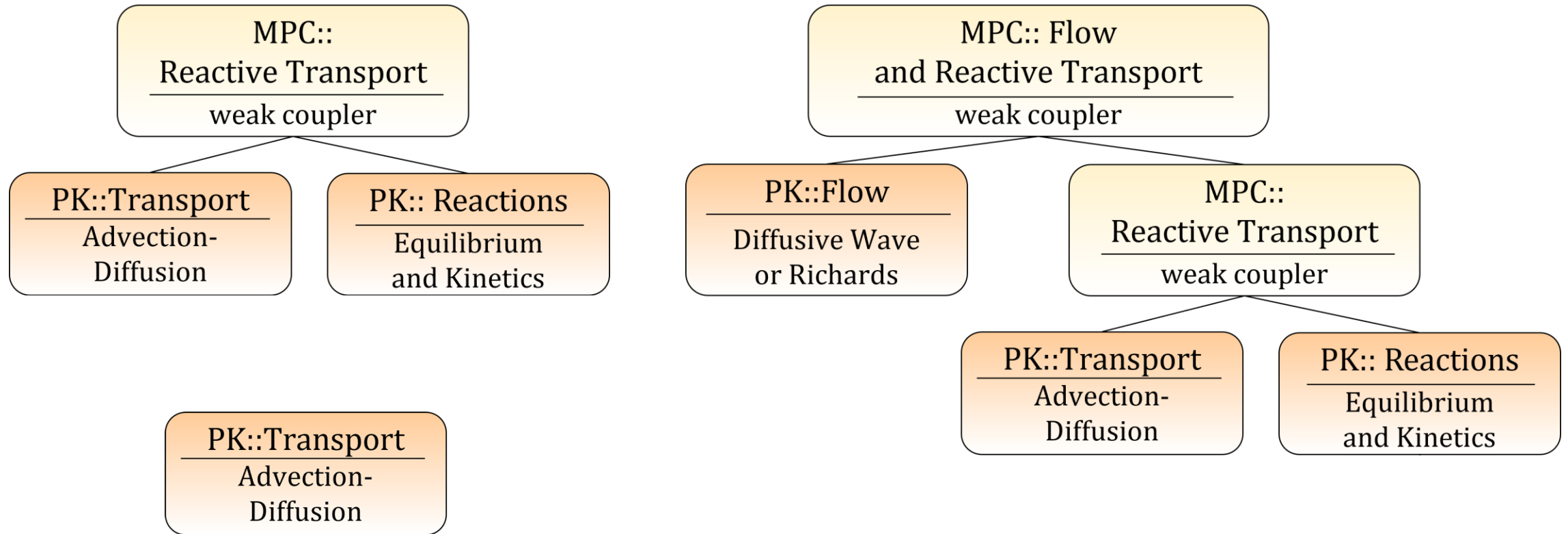


Reactive Transport with ATS





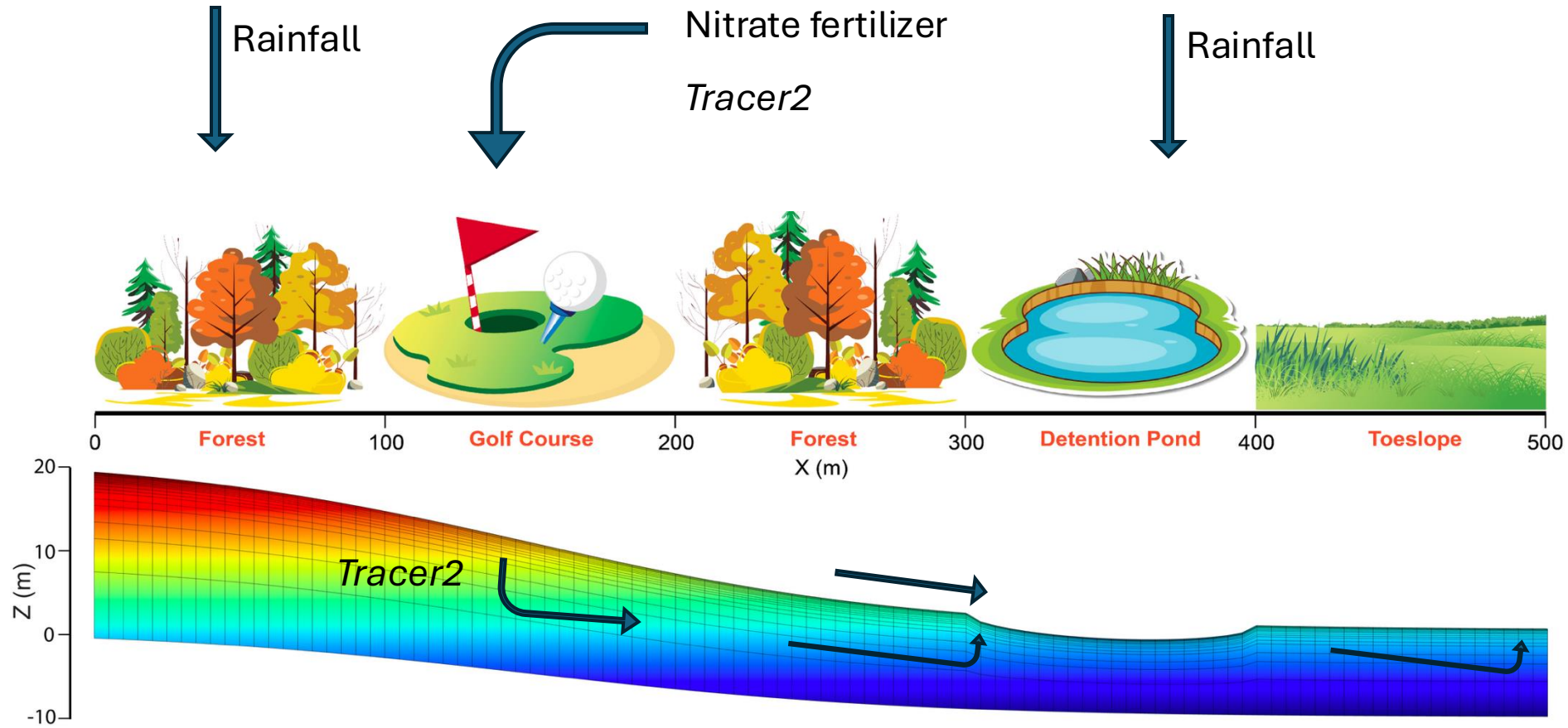
Reactive Transport with ATS





Exercise 1

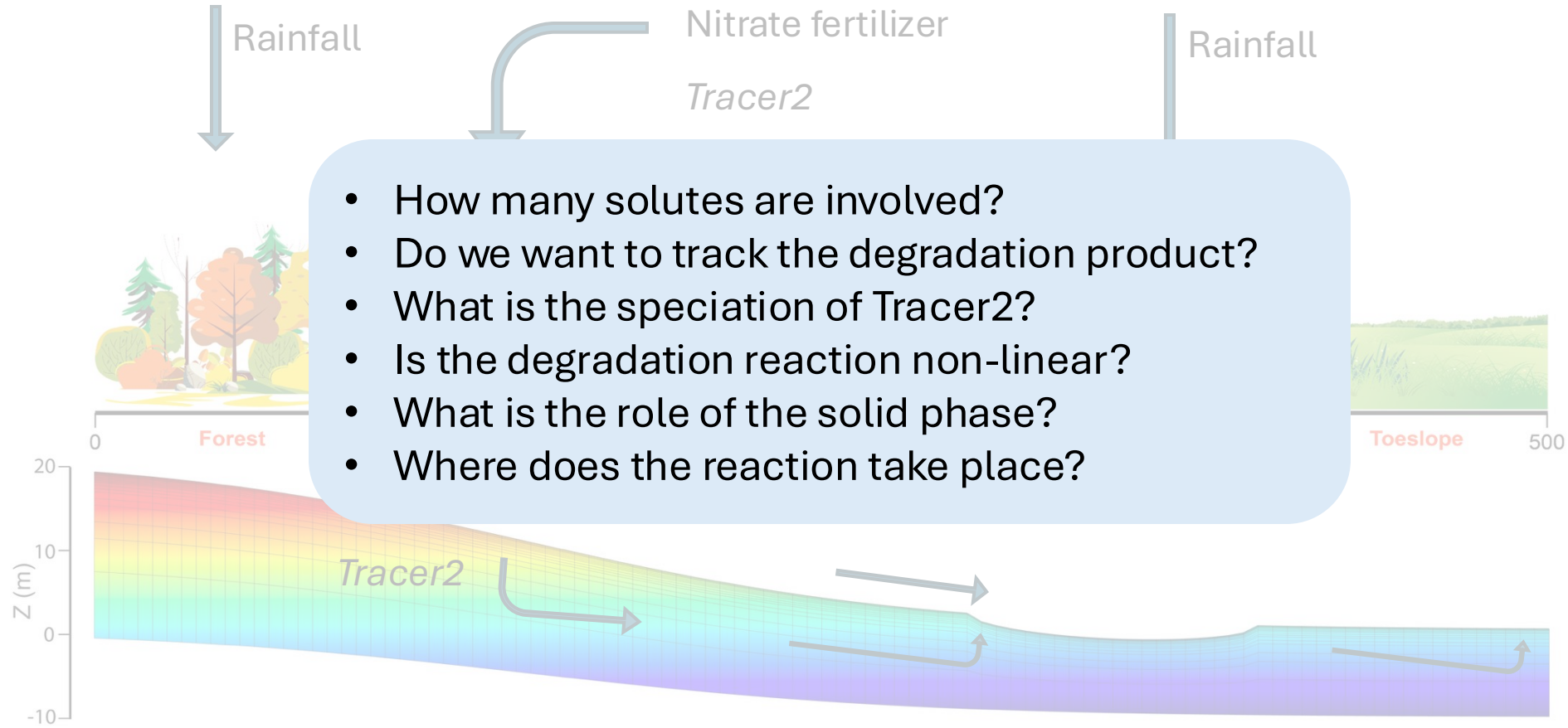
Tracer2 degrades as it moves in the environment according to a first-order rate: $r = -kc$





Exercise 1

Tracer2 degrades as it moves in the environment according to a first-order rate: $r = -kc$





Exercise 1

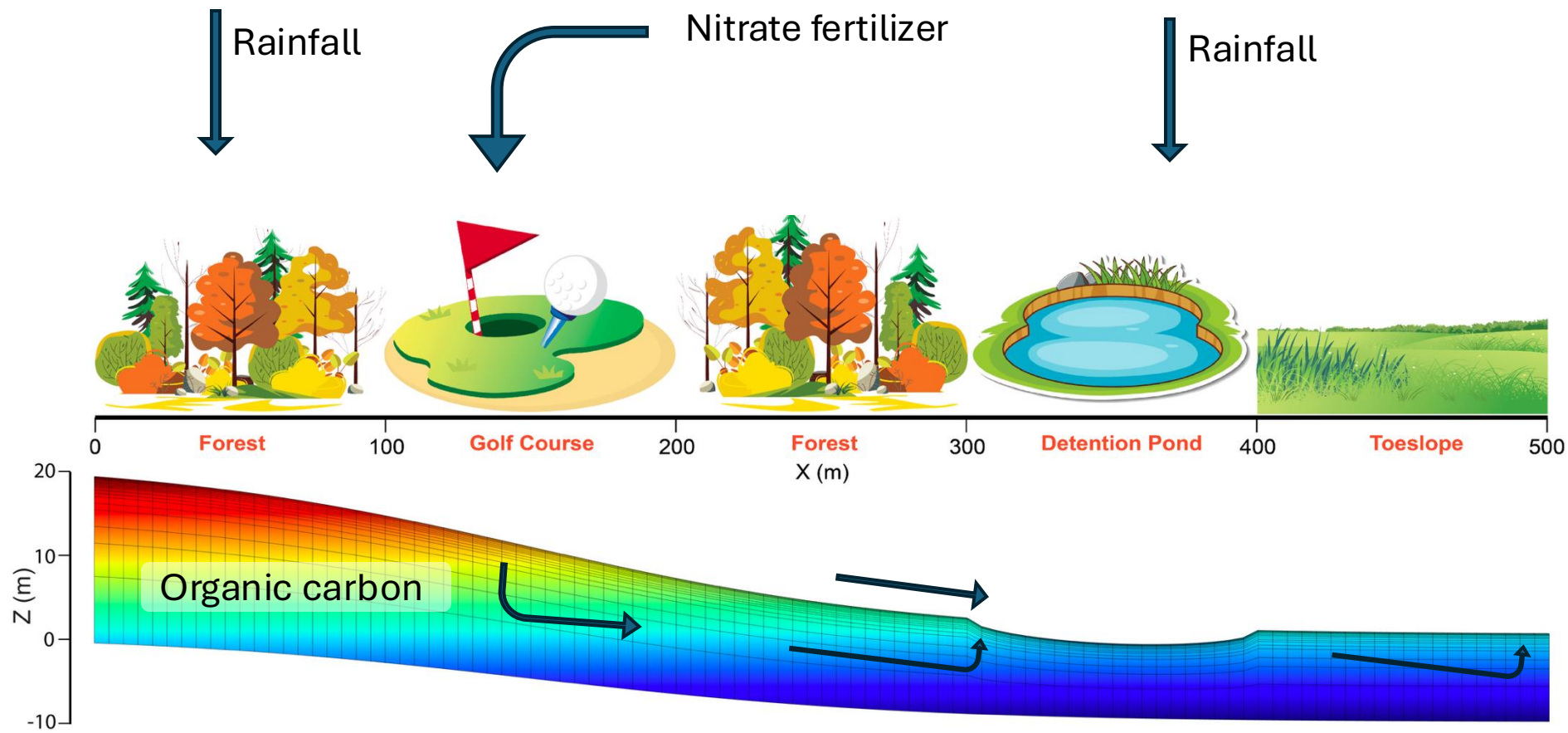
Tracer2 of the integrated hillslope transport problem from the previous session degrades as it moves in the environment according to a first-order rate:

$$r = -kc$$

- How many solutes are involved?
- Do we want to track the degradation product?
- What is the speciation of Tracer2?
- Is the degradation reaction non-linear?
- What is the role of the solid phase?
- Where does the reaction take place?

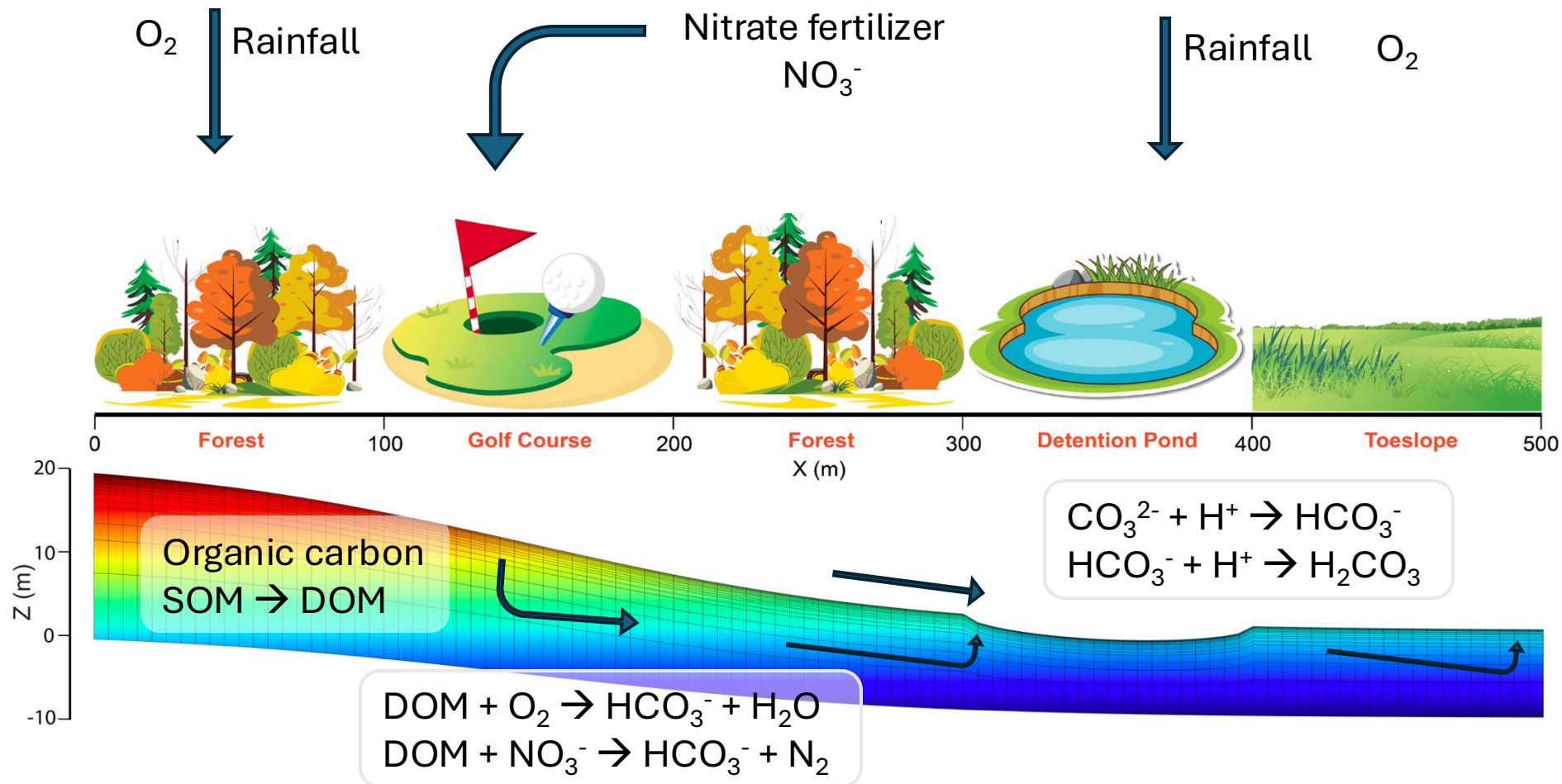


Multicomponent geochemistry with ATS



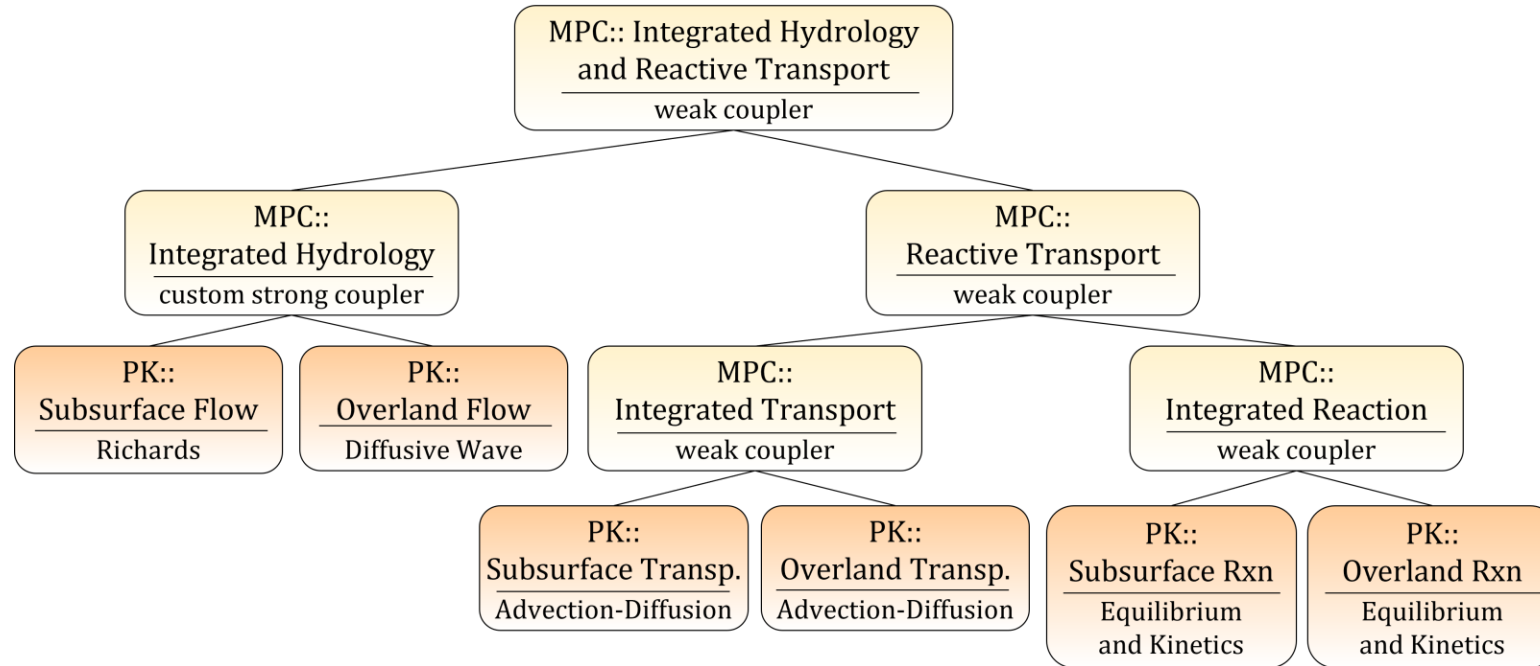


Multicomponent geochemistry with ATS



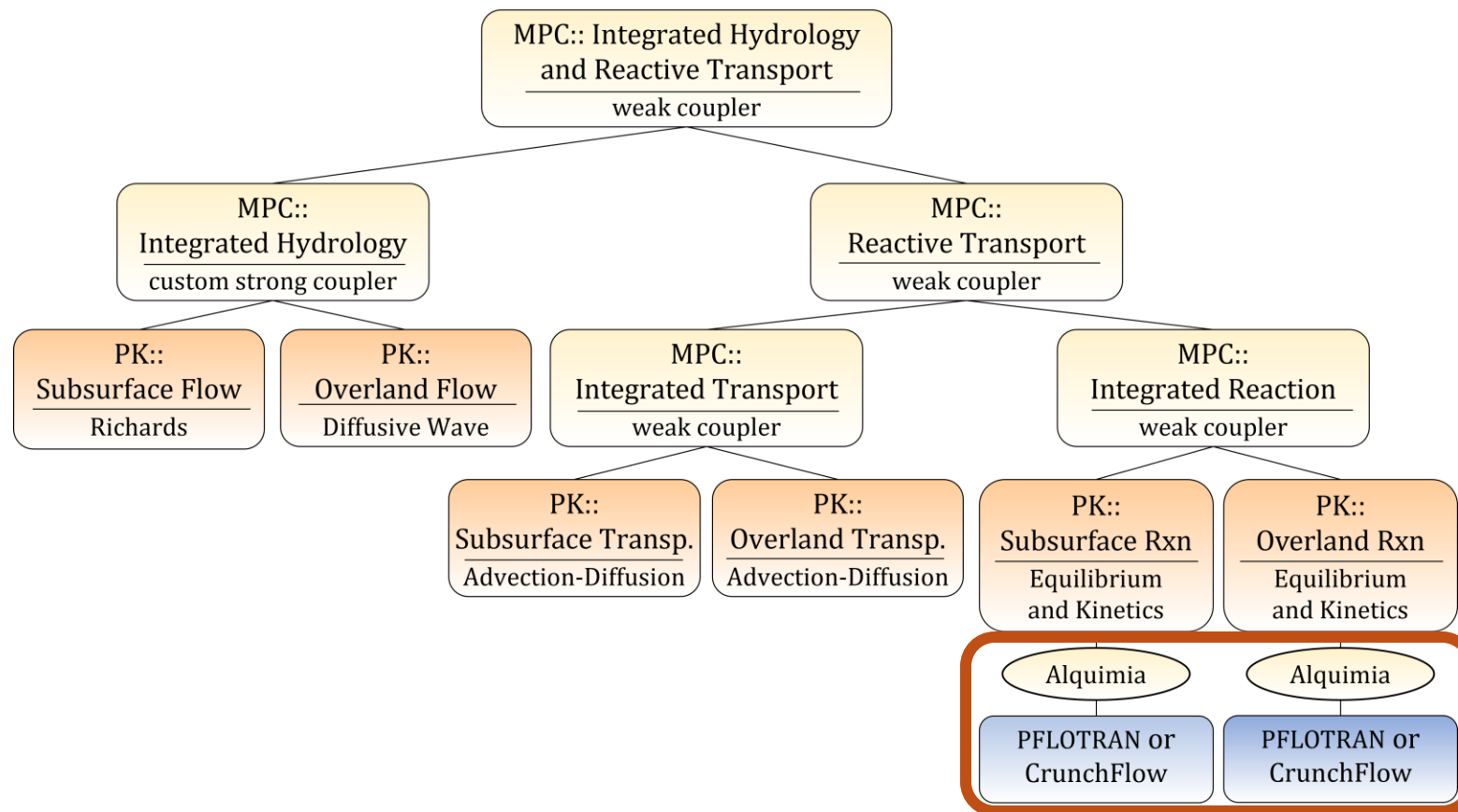


Multicomponent geochemistry with ATS





Multicomponent geochemistry with ATS





Multicomponent geochemistry with ATS

Water Resources Research®



METHOD

10.1029/2022WR032074

Key Points:

- A new reactive transport model is built on an existing integrated hydrology code that couples surface-subsurface flow implicitly
- Integrated transport is solved explicitly using a novel coupling approach
- Multicomponent reactions are implemented in an interoperable manner through a common interface to external geochemistry codes

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Citation:

Molins, S., Svyatsky, D., Xu, Z., Coon, E. T., & Moulton, J. D. (2022). A multicomponent reactive transport model for integrated surface-subsurface hydrology problems. *Water Resources Research*, 58, e2022WR032074. <https://doi.org/10.1029/2022WR032074>

A Multicomponent Reactive Transport Model for Integrated Surface-Subsurface Hydrology Problems

Sergi Molins¹ , Daniil Svyatsky² , Zexuan Xu¹ , Ethan T. Coon³ , and J. David Moulton²

¹Lawrence Berkeley National Laboratory, Berkeley, CA, USA, ²Los Alamos National Laboratory, Los Alamos, NM, USA,

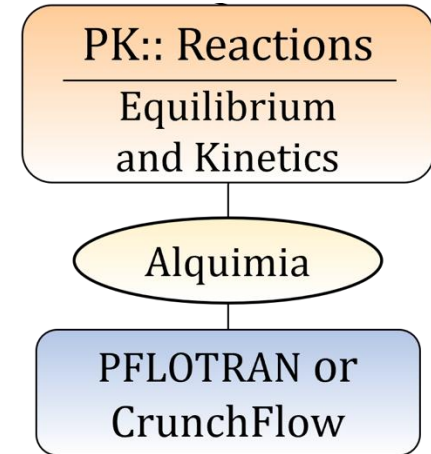
³Oak Ridge National Laboratory, Oak Ridge, TN, USA

Abstract Despite the widespread use of integrated hydrology models in a variety of applications, consideration of multicomponent reactive transport is still not common. The implementation of these processes requires coupling transport at the surface-subsurface interface and efficient solution of the non-linear geochemical model that is consistent with the integrated hydrology solution. The Advanced Terrestrial Simulator provides a flexible multiphysics framework that facilitated this process. In this work, the integrated reactive transport process kernel (PK) was weakly coupled to the integrated hydrology PK. In turn, integrated transport and reactions were coupled using an operator splitting approach. This splitting enabled an explicit solution of the integrated transport problem, including a novel algorithm to calculate exchange fluxes across the surface-subsurface interface and a point-by-point solution of the geochemical problem. Geochemical capabilities were added using well-established external codes, but rather than using a custom interface to each, a generic interface was used that clearly specifies the variables and operations used by the chemistry PK. The implementation is demonstrated with two example simulations: transport of a tracer in a soil column as it saturates over time and water ponds on the surface and reactive transport in a hillslope driven by successive wet-dry cycles that result in infiltration, runoff and exfiltration processes.



Multicomponent geochemistry with ATS

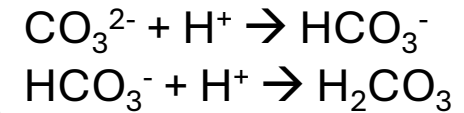
- The transport problem is solved by ATS for geochemical components (total concentrations.)
- The geochemical problem is set up and solved by either PFLOTRAN or CrunchFlow grid cell by grid cell.
- The geochemical solution includes the concentrations of all species, mineral volume fractions, pH, etc.
- The solution of the geochemical problem updates the total concentrations in ATS.
- The total concentrations in mol/L are converted to mole fractions (both available in output file.)





Multicomponent geochemistry with ATS

$$\left\{ \begin{array}{l} m_j = (K_j \gamma_j)^{-1} \prod_{i=1}^{N_c} (\gamma_i m_i)^{\xi_{ij}} \quad j = 1, \dots, N_x \\ \chi_i = \frac{\rho}{\eta} \left(m_i + \sum_{j=1}^{N_x} \xi_{ij} c_j \right) \end{array} \right.$$



Total Inorganic Carbon

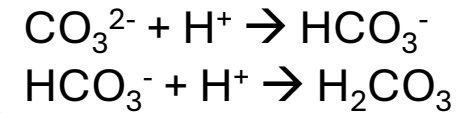
$$\text{TIC} = \text{HCO}_3^- + \text{CO}_3^{2-} + \text{H}_2\text{CO}_3$$

$$\frac{\partial \Theta_g \chi_i}{\partial t} - \nabla \cdot (\phi S_w D_g^* \nabla \eta \chi_i) + \nabla \cdot (\mathbf{q}_g \chi_i) = Q_g \chi_{g,i} + R_{g,i}$$



Multicomponent geochemistry with ATS

$$\left\{ \begin{array}{l} m_j = (K_j \gamma_j)^{-1} \prod_{i=1}^{N_c} (\gamma_i m_i)^{\xi_{ij}} \quad j = 1, \dots, N_x \\ \chi_i = \frac{\rho}{\eta} \left(m_i + \sum_{j=1}^{N_x} \xi_{ij} c_j \right) \end{array} \right.$$



Total Inorganic Carbon

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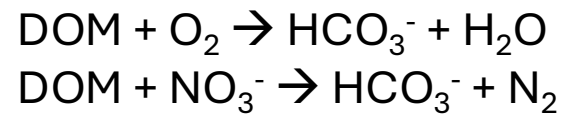


Multicomponent geochemistry with ATS

$$\frac{\partial \Theta_g \chi_i}{\partial t} - \nabla \cdot (\phi S_w D_g^* \nabla \eta \chi_i) + \nabla \cdot (\mathbf{q}_g \chi_i) = Q_g \chi_{g,i} + R_{g,i}$$

$$R_{p,i} = f(m_i)$$

Organic carbon
SOM $\rightarrow$ DOM

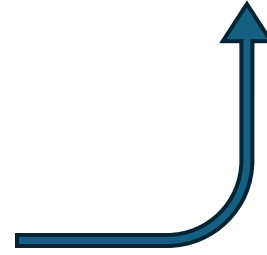




Multicomponent geochemistry with ATS

$$\frac{\partial \Theta_g \chi_i}{\partial t} - \nabla \cdot (\phi S_w D_g^* \nabla \eta \chi_i) + \nabla \cdot (\mathbf{q}_g \chi_i) = Q_g \chi_{g,i} + R_{g,i}$$

$$R_{p,i} = f(m_i)$$



$$\left\{ \begin{array}{l} R_{calcite} = A_{calcite} \left(k_1 a_{H^+} + k_2 a_{H_2CO_3^*} + k_3 \right) (1 - Q/K_{calcite}) \\ \frac{\partial \Phi_{m,p}}{\partial t} = f(m_i) \end{array} \right.$$



ATS input

Define overall coupled problem with PK tree

Chemistry PK: specify engine and its input file name,
specify number and name of components

Transport PK: specify number and name of
components, assign solution to initial & boundary
conditions, and sources

Specify and write output

Engine input

Define Component and Secondary species

Define Mineral, Exchange and Sorption sites

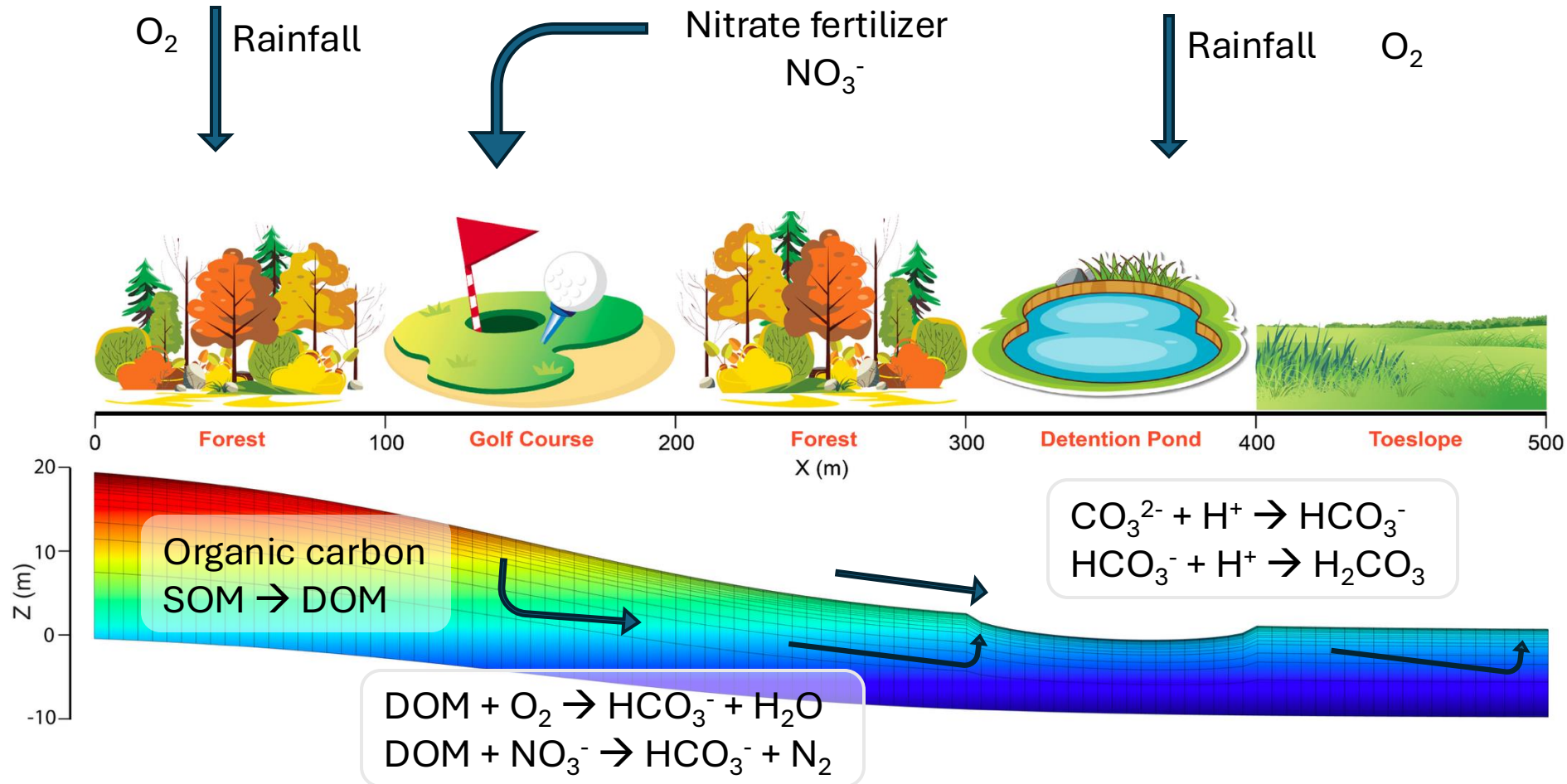
Define kinetic rate expressions

Compute geochemical solutions
defining initial and boundary conditions



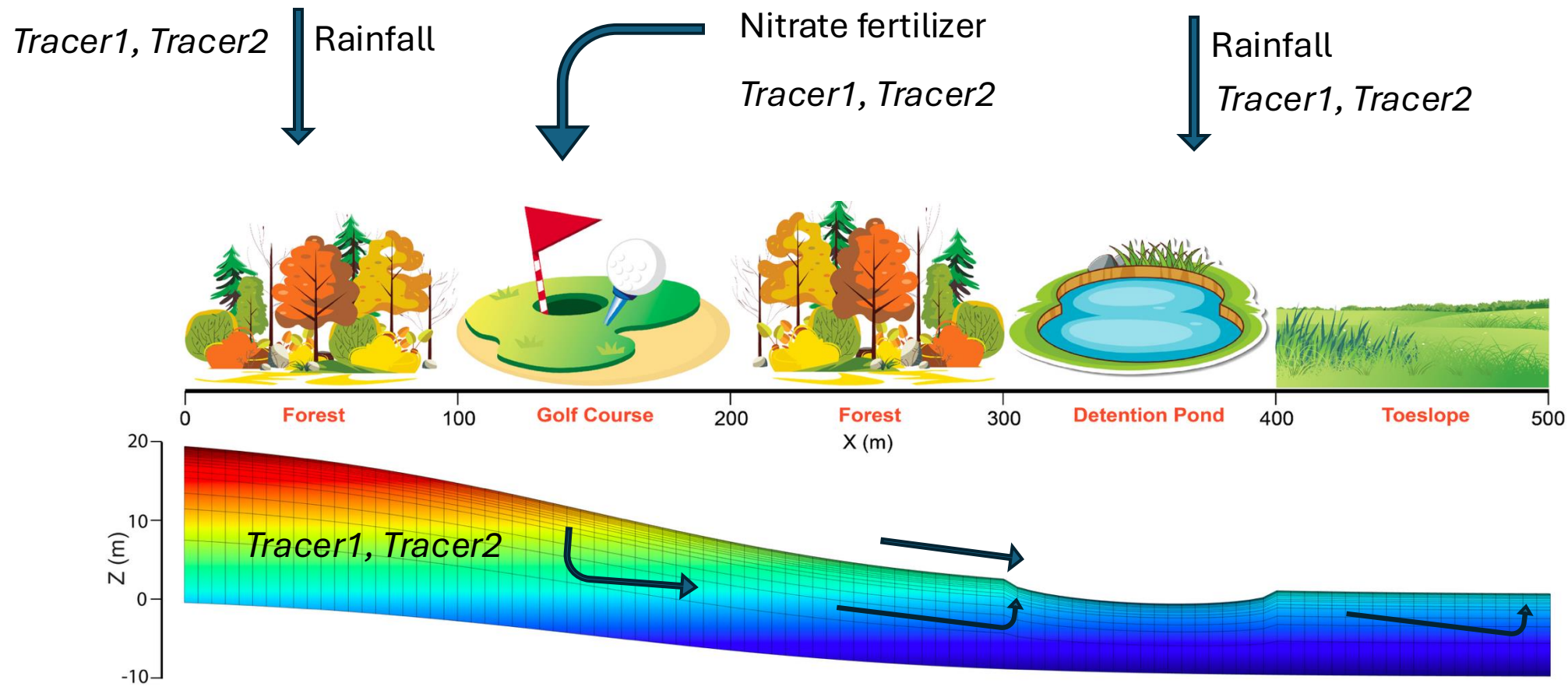
Exercise 2

Use of nitrate fertilizer on the golf course drives redox processes that use organic carbon as electron donor



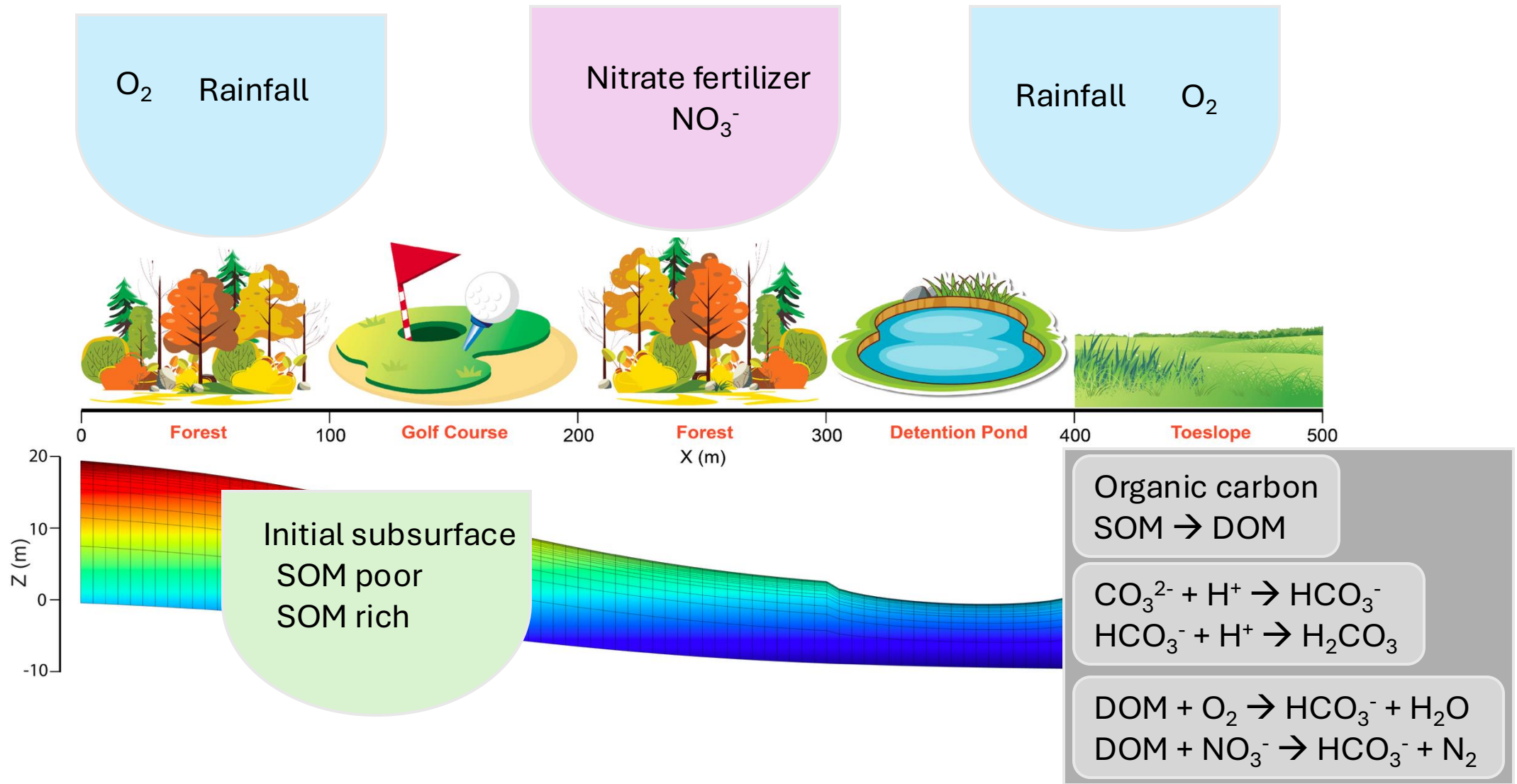


Exercise 2a





Exercise 2b





Exercise 2c

